

Distal Extremity Fracture Stabilization

Authors:

Rebecca Kestin*, Chetan Sharma*, Austin Brown*,
Thanh Nguyen*, Rachel Hoffman, Dr. Jay Connor,
Professor Alex Slocum

*Indicates authors contributed equally to paper

Abstract

This paper presents the design for a highly compactable splint for lower extremity fractures and sprains, and the analysis on the experimental results. The splint is designed for EMTs serving cities who need a compact, versatile, and durable splint that can be applied quickly. Through the use of an origami-style matrix of folding panels, we are able to achieve a highly compact design (8"x8"x1" when stored). The fabric hinges are designed to be durable enough to last tens of thousands of folding and unfolding cycles and to provide shear engagement between the panels. In order to verify the strength of the splint, finite element and analytical models of the splint were derived and the models were compared to the experimental strength of the splint. The FEA model proved to be accurate in terms of trends and within a factor of 1.5 numerically. The analytical model had an error under 20%. The experimental results and feedback received from EMTs were overwhelmingly positive and suggest that the splint can be easily altered to sufficiently satisfy the primary functional requirements. However, improvements are needed to achieve a better ergonomic fit on a wider variety of limb sizes.

Introduction

A fracture presents immediate hazards to a patient. Jagged bone edges are often created when a limb is fractured; if the fracture is moved, these edges can lacerate surrounding tissue or blood vessels, causing immense pain or serious damage. The patient is met at the site of the injury by the EMT. EMTs use splints to immobilize the limb of the patient and prevent further injury before transporting them to the ambulance. The EMT drops off the patient at the hospital. In the hospital the limb is x-rayed, often before the splint is removed. Therefore, the splint must be made of x-ray translucent materials.

In addition to immobilization of the limb and x-ray translucency, modern splints must satisfy additional requirements. Splints must be able to be applied quickly in the field and removed quickly in the hospital. EMTs traditionally spend less than 5-10 minutes on the scene when dealing with a fractured limb. If a splint is too complicated to remove, hospital personnel will sometimes cut the splint off, making it no longer reusable. EMTs have extremely limited space in the ambulance, meaning any splints must be compact. The same splint must be able to be used for multiple sizes of limbs or different limbs. This is particularly relevant in countries such as the United States where obesity is on the rise. Splints must also be light. EMTs often do not know whether a splint will be needed for a given call, and hence carry it from the ambulance to the location of the victim on many calls throughout the day even when they do not ultimately apply it. Reusable splints must be durable and easily cleanable.¹⁻⁵

This paper proposes creating such a splint by means of an "origami" splint, which folds to take up less space. This splint is made of discrete padded panels held together by fabric and metal hinges. When folded flat, the panel takes minimal space. When unfolded and Velcro strapped around a leg, the splint gains rigidity from an increased moment of area and provides support for the fracture.

Background

A fracture is defined as a complete break in a bone. These breaks are often sustained through trauma; contributing factors may include old age or underlying health conditions. A fracture will often result in jagged bone ends and create sharp fragments. If these bone ends pierce the skin, the fracture is referred to as a compound fracture.



Figure 1: X-ray Image of fractured Tibula and Fibula.
[a]

If left as is, a fracture is likely to cause the patient extreme pain when disturbed; the jagged bone ends can lacerate internals. A splint serves to prevent this by holding the fracture in a fixed position. This allows the patient to be moved without causing further harm.

In the field, it is impossible for EMTs to diagnose a fracture. When an EMT arrives on the scene and suspects a fracture or serious sprain has occurred, they will typically splint the limb to stymie the risk of further injury. This allows them to load the patient into an ambulance without risking additional injuries. The suspected fracture is later X-rayed when the patient arrives at a hospital.¹⁻⁵

EMTs do not wait at the hospital to retrieve the splint. If the splint is reusable, the hospital will store it after removing it from the patient, and the EMT will pick it up at a later visit to the hospital. This means that if the EMTs use reusable splints, they will need to own many more splints than they have in their truck at any given time, which can impose a significant financial burden on smaller groups.

Additionally, EMTs do not have much space to carry equipment in their ambulance or on themselves to the scene of injury. EMTs must serve people of various shape and sizes, they have to carry splints of different sizes to accommodate for all the possible patients.

Functional Requirements

Conversations with EMTs led to the realization that the functional requirements for splinting vary wildly for each EMS group. EMS groups from major cities splint multiple patients a day and prioritize reusability and speed of use above all else. Groups from smaller cities and schools rarely encounter injuries beyond simple sprains and therefore prioritize comfortable and versatile splints with a low price point.

A number of requirements were still found to be common to both groups. Most importantly, the splint must immobilize the limb for transport to and within the hospital. Immobilization reduces the patients pain and prevents further damage to a fracture. It also prevents the jagged ends of fractured bones from cutting nearby tissues. Whilst not directly quantifiable, most EMTs agreed that a splint that felt “snug” would fit this requirement. Another commonly used definition is that a limb must be immobilized both above and below a fracture.

An additional requirement common to both groups was comfort. A splint that put a patient in discomfort would defeat its purpose; for this reason, EMTs preferred splints that conformed to the broken limb. EMS groups in rural areas can spend up to 45 minutes transporting a patient to a hospital and therefore value having a more padded splint. For EMTs in cities,

the trip often lasts under 10 minutes and having a well-padded splint is less important.

Size and versatility were also required by most groups. Due to the wide variety of equipment EMT’s need, they often have little storage space available for splints. Often, EMTs use their backpacks to carry gear from the ambulance to the patient. A splint design that is small enough to fit inside a backpack with room for other equipment is valuable. EMTs often need to carry splints of multiple sizes. Patient sizes vary significantly from small children to obese adults and legs are often thicker than arms. A single splint that conform to a wide range of sizes reduces the number of splints an EMT must carry with them.

Some EMS groups highly prioritized a splint that is able to function even with rough treatment. EMTs from Boston stated that they use the Manny splint because it can be dragged repeatedly through a construction zone without being damaged. EMS groups that highly valued durability also tended to stay away from vacuum and inflatable splints. If a valve was damaged or the splint popped, it would be completely unusable. Similarly, EMTs emphasized splints that consist of as few parts as possible - preferably one - as additional parts could get lost.¹⁻⁵

According to the EMT handbook, limbs should always be splinted in place. This means that if a limb is at an odd angle when the EMT arrives at a scene they should transport the patient with the limb still at that odd angle (this does not apply to femur fractures requiring traction splints). The EMT should not try to set the limb at the scene because they do not have the training to do so safely. Furthermore, EMTs do not carry the necessary equipment to do so. Most splints can accommodate this requirement. The Manny Splint cannot.⁶

A final requirement common to all groups was X-ray translucency. Patients are often X-rayed before the splint is removed.¹⁻⁵

The Origami Splint targets EMS groups that value durability and compactibility most highly but also want a reasonably versatile and fast splint. Consequently, it works well for EMS groups located in cities. It is more versatile and compact than the Manny splint; it is more durable and faster to apply than vacuum or inflatable splints; it is also cheaper, more compact and faster to apply than the REEL splint. It is less desirable for EMTs in rural areas. It has less padding than many splints, and would not work as well for hour-long ambulance rides to a hospital. For compound fractures in which the limb is bent at an odd angle, the splint must be reconfigured and therefore takes more time to apply.

Prior Art

There is no gold standard when it comes to splinting. Due to the wide range of priorities each EMS department has, there is no one perfect splint. Different EMS groups value price, size, durability, cleanability, versatility, ease of use, and patient comfort to different extents. We aim to outline the benefits and drawbacks of the most commonly used splints below.

Most splints consist of two main parts: a stiff member that the limb can be attached to in order to remain straight and padding to cushion the area between the bone and the limb. The simplest such example is called the board splint (\$10). This consists of two wooden or metal rods covered in a thick layer of padding. These boards are put on either side of the injured limb and tied on with fabric cravats. This can be a time consuming process because multiple cravats must be individually threaded under the limb and boards and tied tightly. Additionally, it is often large and unwieldy because the boards must be long enough to accommodate the largest of patients.⁷



Figure 2: A single board splint. This is the simplest type of splint consisting of a board with padding on one side. Two board splints are used together with cravats to splint arms and legs. [b]

A faster, though more expensive, splint is the vacuum splint. A vacuum splint consists of a sealed bag filled with small pellets. This bag can be wrapped around a limb and will conform to the limb's features. If air is evacuated from the bag, the pellets lock together and cause the splint to transition into a rigid state. Whilst effective, these splints cannot function if they are punctured or separated from their air pump.⁸



Figure 3: A vacuum splint. This is a more expensive and complex splint. It conforms well to almost all geometries. The downside of this type of splint is that it is rendered useless if the seal is broken or the valve is damaged.

An alternative, disposable splint on the market is the SAM splint. This \$10 splint consists of a thin roll of aluminum alloy sandwiched between two thin layers of foam. The splint can be cut with scissors to the necessary length. To give it stiffness, the EMT shapes the splint into a T-shaped or U-shaped cross section, increasing the second moment of area. The splint is then tied to the patient using cravats. Some EMTs prefer these splints because they are compact and light and can be easily cut to the right length unlike board splints. However, they can be time consuming because they must be cut, folded, applied, and then tied on.⁹



Figure 4: A SAM splint. This is a cheap and versatile type of splint but slow to apply and rather floppy.

The Manny Splint consists of a series of PVC tubes laid parallel in a durable case. The splint can be rolled around a patient's limb like a sushi mat. This splint is extremely durable and lasts for decades. However, because the splint consists of completely straight tubes, lots of padding must be used to allow it to conform to the patient's limb. This makes the splint big and bulky. Additionally, because the size cannot be adjusted EMTs must carry several sizes of splint in their ambulance at all times.



Figure 5: A Manny Splint. This is the preferred splint of Boston EMS. It is extremely durable, but it is bulky and cannot splint injuries in place.

The REEL splint is a special splint intended for military applications. It is a collapsing metal splint designed to splint multiple types of fractures on both the upper and lower body (including femoral fractures). It is extremely durable and versatile, but it is also unintuitive; many of our interviewed EMTs struggled to operate it. It is also extremely expensive (~\$700).⁸

Image Courtesy of Italeti



Figure 6: A Reel splint. This military grade splint can also accommodate femoral fractures. It can be complicated to operate without prior training. [e]

Many other miscellaneous styles of splint also exist. Some examples include: inflatable splints, telescoping splints, pillow splints, and cardboard splints.

Concepts and Strategy

Our creation of a concept began with the analysis of three broad strategies - a splint made of discrete locking elements, a splint made of a continuous locking media, and a splint that would add additional functionality. A cursory analysis of prior art and feasibility led to us pursuing the first strategy.

We proposed several concepts of this strategy. The first was a splint made up of beads threaded on a cable. When the cable was tightened, the beads would interlock. This principle was inspired by an adjustable micrometer stand. This strategy was proved infeasible by further mathematical analysis. The next concept was a splint made of a number of roller chain segments. Roller chains are very stiff in bending across the rollers, while maintaining the ability to roll. Several roller chains were used, and when folded around the leg, all axis of bending were fixed. This splint worked well however was unfortunately very heavy because of the steel roller chains. It could also suffer from buckling failure when undertaking a heavy load. A splint made of spring steel was also considered, but initial testing revealed it to be too dangerous due to its sharp edges and stored energy.

Our final splinting concept was chosen to be the folding panel splint. Initial tests with a wrist sized model proved it to be highly effective at affixing limbs. Several sketch models were made to find a relatively ergonomic fit; once this was achieved, mathematical modeling was

used to predict the behavior of this type of splint. Material choices and dimensions were made through calculations informed by Instron testing.

Once all design parameters were found, a prototype splint was built and tested for improvements. This prototype was used to guide the design of a final splint with minor fixes and changes.

Analysis

We created two main mathematical models for analyzing the flow of forces through the splint. These models were mainly aimed at discovering how shearing between panels on our origami splint would affect force distributions around the splint.

The first model was a FEA program programmed in Python. This model considered the forces along each joint and adjusted the displacements iteratively to solve for a final force and displacement. The benefits of coding the entire model ourselves was that a great amount of detail could be gleaned from directly reading the joint stresses, and that a nonlinear stress-strain curve could be specified for the fabric joints. An ultimate goal (which was unfortunately never reached) was to run tests on many sizes of panel, to find the minimum panel size for safety factor of three. Minimizing individual panel size would lead to a more ergonomic splint, as it could contour around the leg more ergonomically.

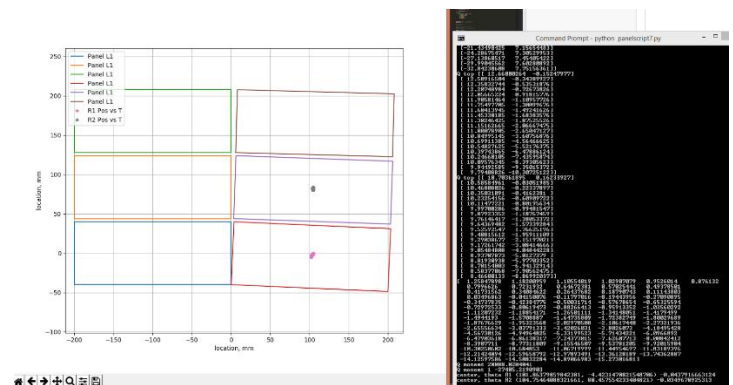


Figure 7: Screenshot of the FEA model running.

This model proved to accurately predict trends in the data, but inaccurate in terms of precise numbers. Therefore, the general trends were condensed to make a simpler spreadsheet model. This spreadsheet model was used to predict our final safety factors.

The second method used for analyzing the internal forces in the splint was derived from fundamental concepts of mechanics. As a panel is loaded, a moment is applied to it. This moment must be counteracted by the fabric connecting it to the adjacent panel. The two panels contact at the bottom of the panels in compression. The fabric that makes up the top of the

hinge expands under tension. Relative to this fabric hinge, the panels can be approximated to be infinitely stiff. The hinge can be modeled as a spring whose spring constant is a function of the relative rotation of the two panels, their initial spacing, and the elasticity of the fabric. In the middle of the hinge (vertically speaking) the panels are not touching and the fabric is not stretching. Consequently, no load is applied. As a result, the panel-hinge assembly exhibits an unusual stress-strain relationship as shown in Figure 8.

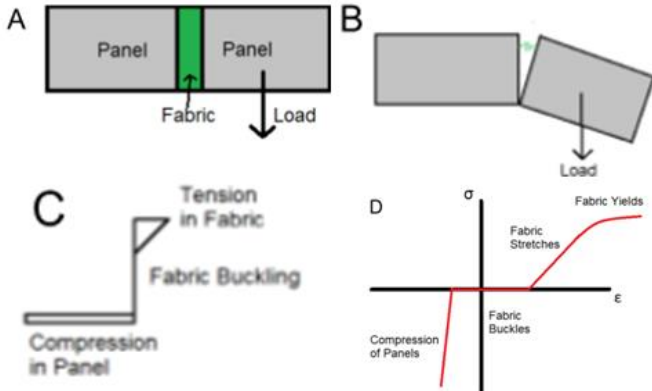


Figure 8: a) loading of splint. b) approximation of fabric as spring. c) visual of compression and tensile loads in splint. d) non-linear stress-strain relationship in hinge.

Based on this geometric understanding, the neutral axis of the entire assembly can be determined. The process for this is similar to the way reinforced concrete structures are analyzed in which the concrete is assumed to take no tensile load. Similarly, the panels only supply compressive loading when in contact, and are modeled as significantly stiffer than the fabric. Next, the bending moment is calculated as a function of the relative angular rotation of the loaded panel to the unloaded panel. Using an iterative process, the calculated bending moment is compared with the applied bending moment. Based on this comparison, a new guess for the relative angle is chosen, and the process is repeated until the error in the guessed moment is sufficiently small.

Device Design

The device was designed keeping both the EMT's and patient perspectives in mind. The device is designed to work with a wide variety of leg sizes, while still preserving patient comfort. It can also be folded into a compact size (from 27" x 16" x .25" to 8"x8"x1") for storage and transportation (Figure 1). This device is made from a series of Lexan panels connected together by Kevlar fabric. The device is protected by an outside layer of vinyl (Figure 2).



Figure 9: Folded Configuration of Origami Splint compared with Manny Splint. As can be seen, the Origami Splint is significantly more compact



Figure 10: Cross section view of the Origami Splint. The splint consists of a matrix of panels held together with Kevlar fabric. Foam is added to increase conformability. A vinyl coating allows the splint to be easily cleaned.

The Lexan panels form a tapered shape when unfolded, to better fit a leg. The device has a zipper running down the middle to allow the splint to be split into two halves (Figure 3). The two halves can be folded into tubes transforming the splint into a board splint (Figure 4). Two slits run from the outside edge of the splint to the inside panels allowing it to fold it up in a right angle (Figure 5). The splint has Velcro straps running down the splint to allow EMTs to easily strap on someone.



Figure 11: The tapered shape allows the splint to conform more easily to the tapered shape of a leg. The zipper allows the splint to be separated into two board splints.



Figure 12: Board Configuration. The splint can be unzipped and rolled into tubes to create two board splints. This allows EMTs to splint angled breaks and knee injuries in place.

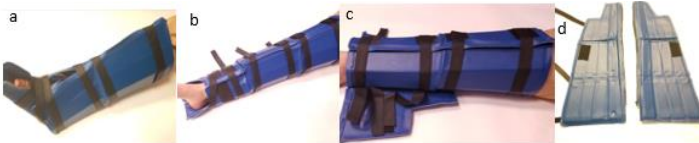


Figure 13: A. Ankle configuration. B. Long tube configuration. C. Short tube configuration. D. Split Configuration

The overall device is significantly smaller and lighter than most splints on the market. The Kevlar hinges allow for repeated folding and unfolding without noticeable wear. The design of the hinges was based off of research into living hinge design.

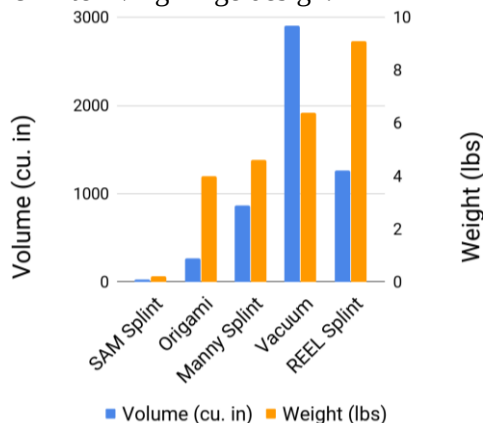


Figure 14: Comparison of splints by size and weight. The origami splint is the smallest reusable splint.

Fabrication

1. All panels were cut from a 24x48 polycarbonate sheet using a waterjet cutter.



Figure 15: First step in assembly process. Lexan panels are cut on a waterjet.

2. Panels were aligned using a board with pegs. A layer of epoxy was added to the panels, and a kevlar fabric sheet was placed on top. The fabric was rotated such that the fibers run crisscross the panels at 45 degree angles. This created a cross-hatched truss-like structure between the panels increasing both stiffness and strength. Care was taken to ensure that no epoxy soaked into the hinge segments of the fabric, keeping the joints soft.



Figure 16: Applying epoxy to Kevlar fabric

3. 1/8" thick foam was cut into the same size and shape as the panels. The foam was then glued on top of the fabric. A layer of vinyl fabric was then glued on the top and bottom of the assembly. The edges of the splint were then sewed together to seal the edge of the splint. The excess fabric was cut off and adhesive was placed in the seam between the vinyl to seal the edges.



Figure 17: Outside Vinyl layer is added

4. Velcro straps were riveted onto the splint. Additional velcro pads were adhered throughout the splint. The splint was then cut in half and had a zipper sewn down the middle.

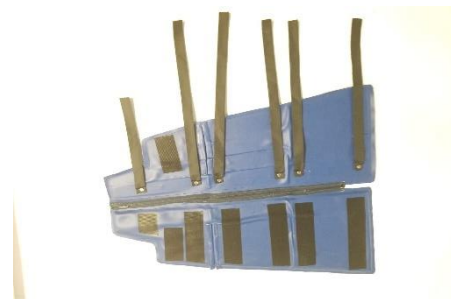


Figure 18: Velcro straps are added to the splint.

Mechanical Testing

Several different tests were run on the Instron machine. The first set of experiments tested the strength of individual joints in either tension or shear. The first samples were made using fiberglass. Two tension samples and two shear samples were tested.

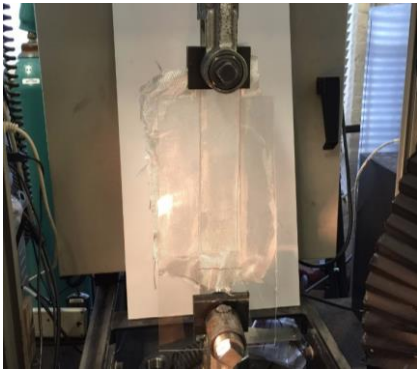


Figure 19: Shear Test Setup

Safety factors for the splint were calculated from the tensile and shear data using an excel spreadsheet. The tension safety factor was calculated to be 4.6, which was more than satisfactory. However, the shear safety factor was found to be only 1.3, which was not satisfactory. Therefore, fiberglass fabric was deemed insufficient for our needs.

Kevlar fabric was selected as a suitable replacement fabric. The Kevlar fabric was also rotated 45 degrees for greater shearing stiffness. This combination worked very well. The safety factor in tension was calculated to be 4.1 and the safety factor in shear was calculated to be 3.1. Both these numbers were satisfactory and therefore we moved forward with Kevlar fabric. The yield stress of the Kevlar was 68.1 MPa.

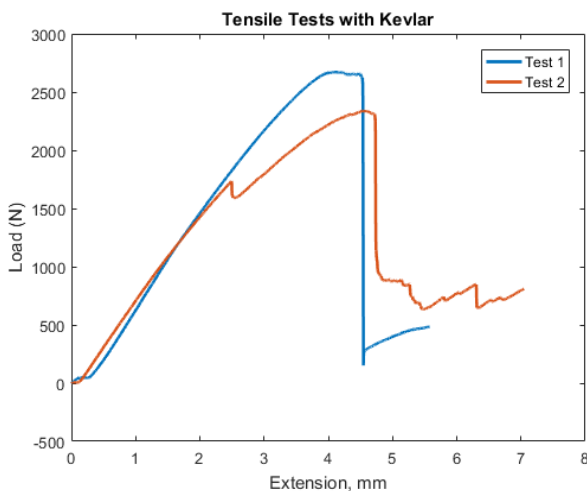


Figure 20: Tensile Strength of Kevlar. The average yield stress was 68.1MPa.

Once Kevlar fabric was selected, we assembled a complete splint to test. The complete splint was tested in bending in the Instron machine. The test proved to be very difficult to execute as the splint was difficult to securely fixture. During the tests the splint slipped a great deal on the form. Therefore, the measured force was accurate, however the displacement data was not. The splint bent in the Instron.

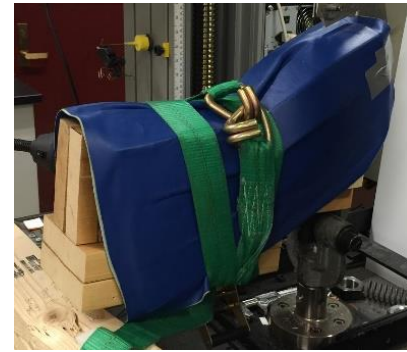


Figure 21: Force vs Displacement Setup for Origami Splint.

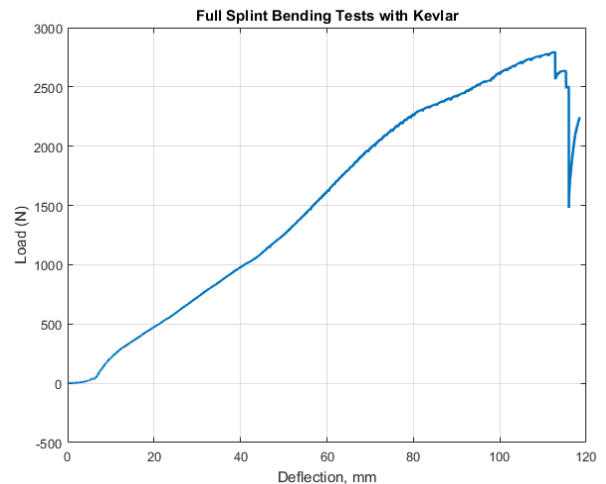


Figure 22: Force/Displacement Data

The splint proved to be very strong, holding 2700N. This is at least 5.8 times our desired load. At this load the splint was very warped. However, after the load was removed, it was still usable of the leg of one of the testers. Upon post-mortem inspection, the Kevlar was found to have sheared along one of the upper panels, precisely where analytical models predicted. This verified at least the trends in our models to be correct.

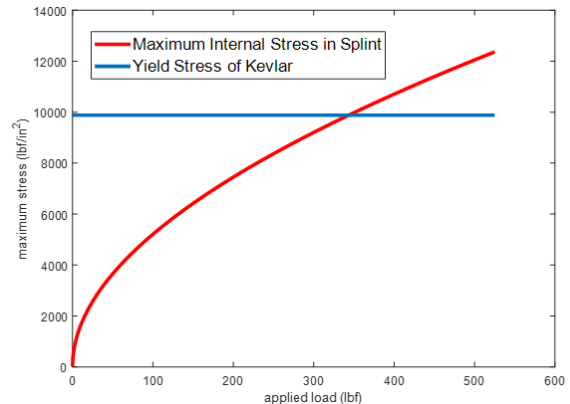


Figure 23: Comparison of expected yield point in splint based on tensile tests and expected internal stress in splint based on analytical Matlab model. As can be seen, the model is a good and conservative predictor of the splint's capacity

Finally, the analytical model was compared with the actual behavior of the splint. Due to the splint slipping slightly relative to the case it was enclosed in, a conversion factor of $\cos(30^\circ)$ is needed to convert from high loads applied by the Instron to loads predicted by the model (which assumes no slippage). The model predicted that the splint would have started to yield at about 5% of the load that was actually applied to it. Given that the internal fabric had begun to show a bit of wear when the splint was examined, this suggests that the analytical model is quite accurate. The model predicts that the splint would begin experiencing minor failure at around 340lbf (392lbf rotated), or 1500N (1746N rotated). The actual splint appeared to have begun yielding around 2200N, although it is impossible to tell because the vinyl was not translucent and hence the onset of failure could not have been visually determined. Given the small sample size and extremely complex geometry, these results (20% error) are quite promising. Furthermore, the model was shown to be conservative.

This analysis is crucial in many respects. Most importantly, a larger splint could be made to serve a particularly obese population, as mathematical analysis is more scalable than hardware. Splints can be scaled to different sizes and adapted to different use cases. For instance, a splint could be constructed for arms or children. Panel size and fabric material could be optimized for cost, weight, size, or other parameters based on few initial measurements and tests. Furthermore, understanding the load capacity is necessary to responsibly make a product for human medical use available on the market.

Human Testing

The Origami Splint was tested by Denny Chen of the Boston EMS unit. Overall he approved of the splint. He was impressed by its versatility. He suggested adding more features onto the splint to fit body parts than just the leg. In particular, if the panel size were decreased, the splint could be unzipped and half of it used to splint an arm or wrist. In addition, the number of velcro straps could be reduced to make the splint faster to apply. Finally, given the current panel size, the splint does not work well on extremely thin limbs.

We also tested the splint on a number of MIT students. The splint was stiff enough, but not very ergonomic. The splint fit well for larger limbs, but did not fit well for the smaller limbs of younger students. The boot feature was found to work well for larger limbs but proved uncomfortable for smaller limbs. The straps also proved unwieldy and non-intuitive. Mechanically the splint proved to be sound, and none of the testers were able to damage it mechanically.

Conclusions & Future Work

Our splint shows promising initial results. It's durable vinyl coating is waterproof. The Velcro and zipper can be washed by hand. It is significantly more portable than the REEL and Manny splints. Unlike the REEL and vacuum splints, it does not have small components that could break off and render the splint unusable. It is faster to apply than any splint except for the Manny splint. However, unlike the Manny splint, it is adaptable to a wide range of fracture types. The Manny splint cannot adapt to bent knee joints or bent fractures. Our splint has a hinge specifically designed to adapt to ankle joints. It can be unzipped and rolled into two board splints to splint angled fractures in place. Initial feedback from medical professionals and Boston EMS has been extremely positive.

Furthermore, mechanical testing on an Instron machine of our splint showed promising results (error less than 20%). In addition, both the FEA and analytical models were conservative. In bending, the yield strength of the splint has a factor of safety greater than 5. This all suggests that the splint can successfully immobilize limbs above and below a fracture point.

Despite these promising results, some issues must still be addressed. The current iteration of the splint has difficulty stabilizing small legs, but works well for large limbs. The splint was designed with the obesity epidemic in America in mind, and hence for larger leg sizes. It is designed to be able to successfully wrap around the leg of a 500lb person and be therefore can be used by more than 99% of the population. However, because it can work so well on larger legs, in the splint's current iteration it has trouble conforming to extremely small limbs.

In future iterations, the splint can be adapted to fit around smaller legs while maintaining its utility for obese patients. The minimum diameter leg that the splint can conform to is determined by the two center panels. These panels take the brunt of the compression load in the splint, and therefore are constructed using stiffer hinges rather than fabric hinges. They are also directly adjacent to the zipper. Consequently, these two panels cannot easily bend around a leg. For future iterations, the panel size can be reduced to allow the splint to conform to smaller legs.

Future testing is also needed to determine the minimum panel size that can be used. The larger the panel size, the less stiff the splint is and the less area the epoxy has to bond with each panel. Furthermore, because the hinge sizes remain constant, the smaller the panels the smaller the panel to hinge ratio. Similarly, more testing is needed to determine the optimal hinge size.

One other weakness of the current Origami Splint is its stiff hinges, requiring the user to work hard

to fold the panels. This is caused by insufficient spacing between the panels. In order to determine the ideal panel spacing, additional testing must be performed. Wider hinges also have less shear engagement, and therefore lead to a weaker splint. However, given the current strength of the splint, the hinges can be made significantly larger without sacrificing splint strength. There is already significant research that has been done in the field of folding of thick joints, but further analysis will be needed to find a joint configuration that fits the unique geometry of the origami splint. Due to the compressible foam panels and potential for stiff hinges if they wick epoxy, current models are not quite sufficient.¹²

In the future, more durability testing should be performed. Due to time constraints, no fatigue loading could be performed on the splint. Fatigue loading of the kevlar hinges should be done to determine the expected lifetime of the splint. A rig can be made to automate the testing of the durability by repeatedly folding and unfolding the hinge. Another test that must be done is a waterproofing test. Ideally, the splint should be tightly sealed enough to be submerged in water without water leaking in.

Even without having done these additional tests yet, our splint shows great potential. Furthermore, by approaching the design process from an analytical perspective and developing models of the strength and stiffness of the splint, it is possible to scale the splint easily. For instance, minimal additional work would be needed to convert the splint to function for children or arms. The splint warrants further development to realize its full potential. In conclusion, the compact nature of the origami panels makes it an attractive choice for city-based EMS groups.

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